MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY, ALLAHABAD

DATA STRUCTURE LAB

ASSIGNMENT- 3

1. An e- commerce company receives a large number of customer orders every day. Each order is identified by a unique Order ID. Before sending orders to the warehouse for packing, the system must arrange all Order IDs in ascending order so that the packing and delivery process becomes easier. Write a C program to implement **Merge Sort** to sort a large-sized array of Order IDs and also find the total number of comparisons performed during sorting and the time required to sort the array.

2. Write a C program to implement **Quick sort** in a given array of larger size.
   - You have to find the number of comparisons and time required to sort the array.
   - What if array elements are already arranged in sorted order? How will you this problem?